Exposure Parameters

Lithography Training Course

Agenda

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Rules of Thumb for Reaching

- Low Feature Sizes
- Good Pattern Placement Accuracy
- Line Edge Smoothness
- Low Overall Exposure Time
- Good Stitching Results
- Good Results on thick Resists
- Optimised Exposure Parameters

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Low Feature Sizes

- High Beam Energy (keV)
  - Depends on substrate/resist system
  - e.g. RAITH150 Two: 20 keV (PMMA on Si)
- Low beam current/small aperture
  - Less Coulomb interaction
  - Large depth of focus
  - Increased exposure time
  - Difficult to focus (low signal-to-noise ratio)

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Low Feature Sizes

- Small write fields
  - » scan amplifier with better signal-to-noise ratio
- Get max. WF for a certain amplifier by increasing WD or reducing kV
- Small step size (< 1/5 of the min. feature size)
- Perfect focusing (check by carbon spot)

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Good Pattern Placement Accuracy

- Use line mode
- Proper settling time – typical values:
  - perfect = 5 ms
  - good = 2 ms
  - suiting = 1 ms
  - poor < 1 ms
- Reasonable beam speed (over sample surface)

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Good Pattern Placement Accuracy

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- Reduce beam speed
  - » decrease current by factor x
  - » increase area step size by factor x
  - » increase dose by factor x

  } → v will be decreased by factor x
- Small write fields → high DAC resolution

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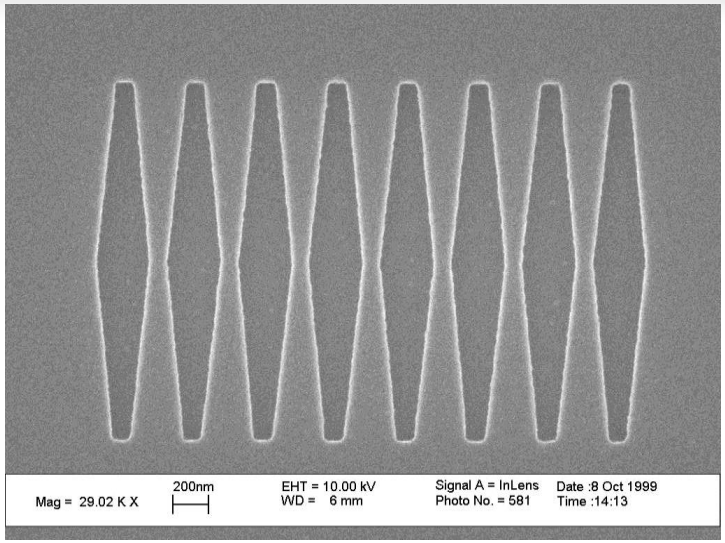
Line Edge Smoothness

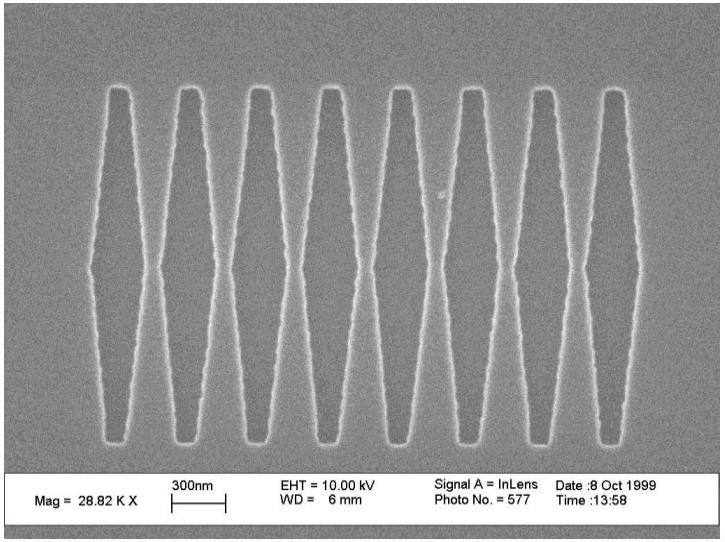
For smooth edges, use

- Line mode
- Proper settling time
- Good focus
- Small write fields
- Small step size

Step size
5 nm



Step size
9 nm



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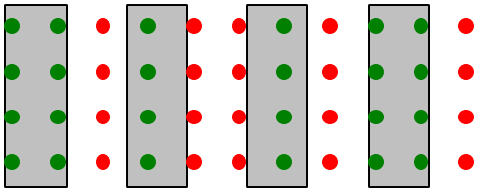
Excursion: Structure Size and Step Size

Important note:

The used exposure step size has to be matched to the design step size in the layout!

Example for exposure of gratings:

Step size (s) does not match grating period (g)



s = 8 nm, g = 10 nm

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Low Overall Exposure Time

- Meander mode
- Low settling time
  - » less placement accuracy (high beam speed)
- High beam current
  - » less placement accuracy (high beam speed)
- Low beam energy (keV)
  - » higher resist sensitivity (lower clearing dose)

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Low Overall Exposure Time

- Large write fields
  - » minimum stage travel
  - » less resolution, more distortion, low signal-to-noise
- Pre-processing of GDSII files
  - » PLS pre-processor
  - » exposure buffer (only for ELPHY IV hardware)

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Good Stitching Results

- Avoid mechanical drift
  - » wait after sample loading until system is stable
- Avoid beam drift
  - » wait after starting the gun: 24h
  - » wait after switching EHT on: 4h
- Take care of good zoom corrections
  - » from 0.9999 to 1.0001 (= 10 nm in 100 µm WF!)
- Perform write field alignment near to exposure field
  - » use carbon spot

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Good Results on Thick Resists

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Critical issue:
Undercut due to forward scattering in resist

In order to avoid undercut…

- Use high keV
- Use resist with high contrast
- Avoid overexposure and overdeveloping
  - » determine clearing dose very carefully
  - » optimize developing process with respect to time and temperature

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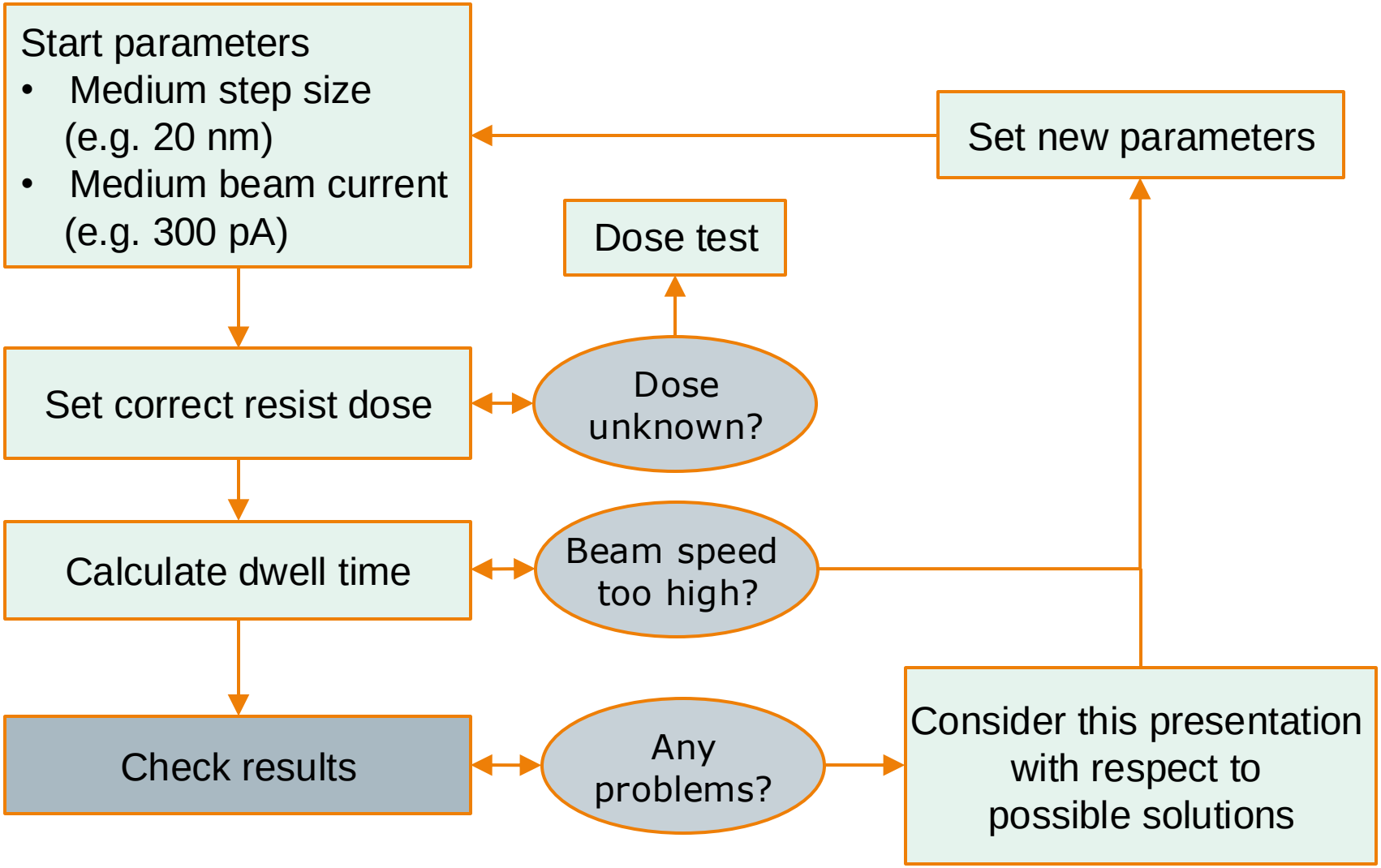
Optimisation of Exposure Parameters



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NANOFABRICATION

Your challenge is our mission.